

Enterprise AI Platform Architecture

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1. Executive Summary & Mission

The Enterprise AI Platform is engineered to deliver reliable, deterministic, and auditable autonomous workflows for enterprise environments. The core challenge in enterprise GenAI deployment is not model capacity, but context precision, hallucination mitigation, and strict enterprise permissioning.

Our unified knowledge retrieval layer powers Forward Deployed AI Engineers and internal systems by combining vector similarity with inverted index keyword search, reranking mechanisms, and citation tracking. This document serves as the reference specification for data ingestion and text chunking requirements.

2. Ingestion Guarantees

To maintain audit compliance across banking, healthcare, and enterprise software customers, the ingestion pipeline must enforce the following guarantees on every processed document:

- **Deterministic Identifiers:** Document IDs must be calculated from cryptographic hashes of source content.
- **Traceable Metadata:** Source document path, page numbers, and detected section headings must accompany every chunk.
- **Lossless Boundaries:** Chunk boundaries must preserve sentence and paragraph coherence rather than performing naive character cuts.
- **Idempotent Execution:** Re-running the pipeline on unchanged source files must produce identical chunk keys.

3. Chunking Strategy & Retrieval Relevance

Traditional naive chunking slices text into fixed token windows (e.g., 500 characters) regardless of document structure. In production enterprise systems, this causes severe degradation in retrieval precision because critical context—such as table headers, clauses in legal contracts, or code blocks—gets severed across chunk boundaries.

We recommend a recursive structural chunking approach. The algorithm attempts to split at structural headers, then paragraphs, then list items, and finally sentence boundaries. Controlled overlap of 15-20% ensures that boundary-crossing thoughts maintain sufficient co-occurrence signal for dense embedding models.

4. Security and Isolation Protocols

Customer documents often contain sensitive Intellectual Property or Personally Identifiable Information (PII). The data plane must adhere to the principle of least privilege:

- Role-Based Access Control (RBAC) must be propagated from the source document down to individual chunk metadata.
- Encryption at rest must utilize AES-256 for all serialized chunk artifacts and vector databases.
- No customer data is transmitted to external endpoints without customer-managed encryption key (CMEK) approval.